Deep Learning on Embedded Systems (esp. NVIDIA Jetson)

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Acronyms

- DL = deep learning
- SVM = support vector machine

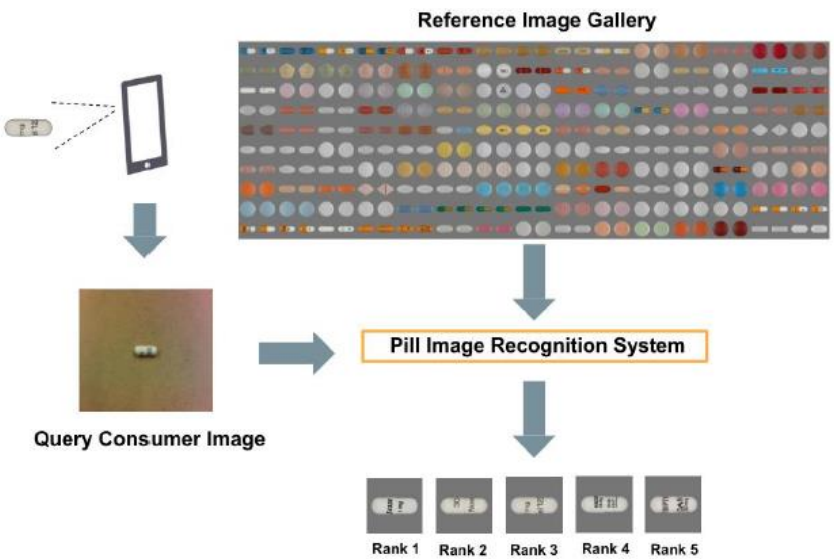
Pill image recognition

Zeng et al., “MobileDeepPill: A Small-Footprint Mobile Deep Learning System for Recognizing Unconstrained Pill Images, MobiSys, 2017

Their entry won first prize in National Library of Medicine pill image recognition challenge

Description of the competition

- Competition: there is a database of high-quality images of pills.
- User takes image of a pill from his mobile phone.
- Task is to find a list of most similar pills to the query image.



Challenge: achieving high accuracy is challenging due to factors such as illumination, phone orientation and existence of a large variety of pills.

Challenges in achieving high accuracy

- Database images have uniform background and plain texture, pill can be localized and segmented using "gradient detection" technique
- However, query images are taken in disparate backgrounds.
- The images may have noise

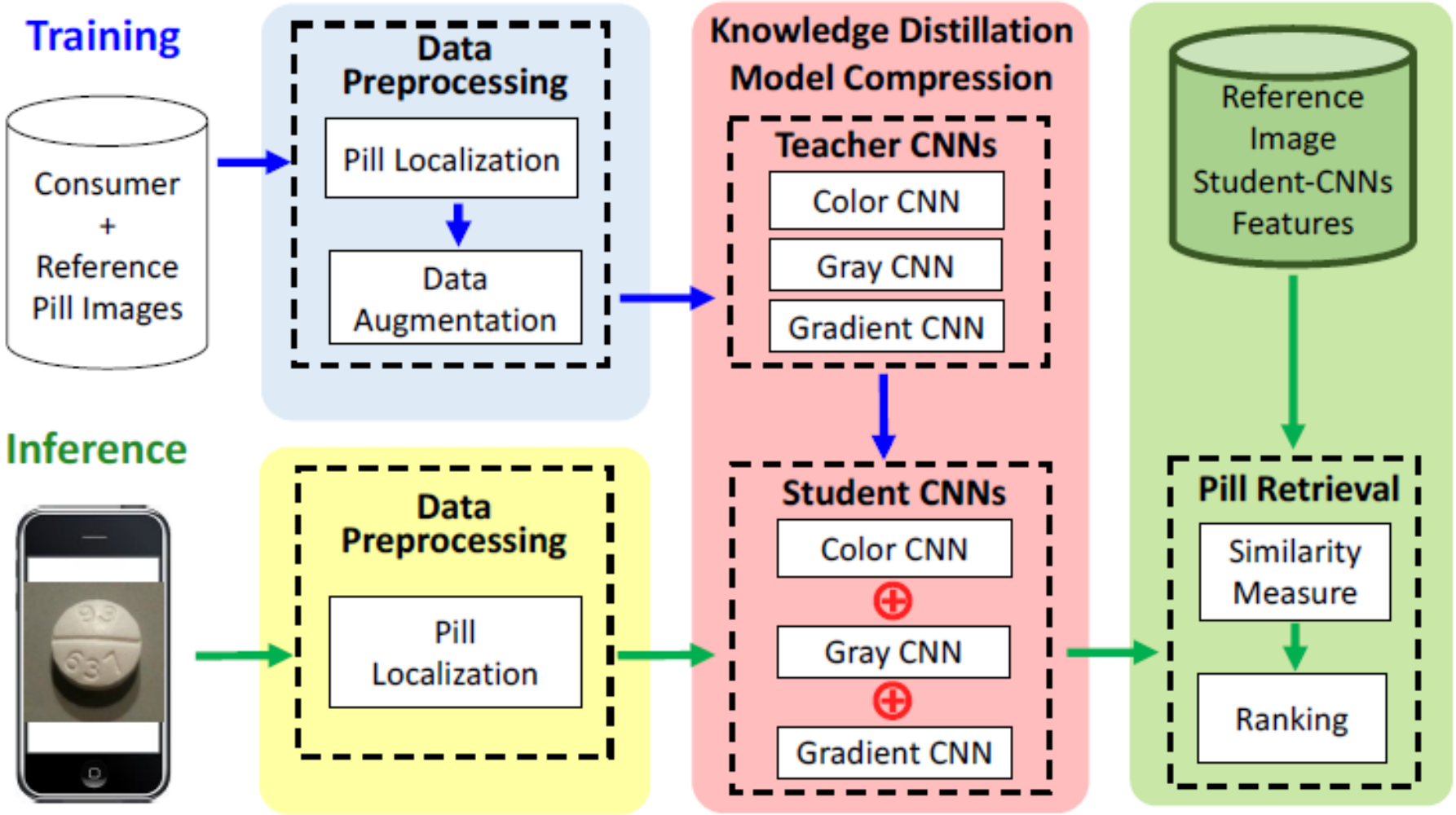
Left : user image
Right: reference image

Achieving high accuracy in detection

- Ideas: Train a pill detection engine using SVM.
- Apply “Gaussian filtering” for simulating blurred images
- Zoom the images to ensure robustness to changes in pill-size
- Apply random translation and rotation to ensure robustness to localization errors

Multi-CNN architecture

- Each pill can be identified by its unique combination of color, shape and imprints
- Challenge: noise may change these, e.g., white pill may appear yellow due to yellow illumination.
- Idea: use a multi-CNN architecture for achieving high accuracy. Generate the gray and gradient images of the original color images to train three independent CNNs.
- CNNs trained on gray and gradient images
  - remove the dependency on color information
  - attain enhanced shape and imprint information for accurately distinguishing different pills


Training
Consumer + Reference Pill Images
Data Preprocessing
Pill Localization
Data Augmentation
Knowledge Distillation Model Compression
Teacher CNNs
Color CNN
Gray CNN
Gradient CNN
Reference Image Student-CNNs Features
Inference
Data Preprocessing
Pill Localization
Student CNNs
Color CNN
Gray CNN
Gradient CNN
Pill Retrieval
Similarity Measure
Ranking

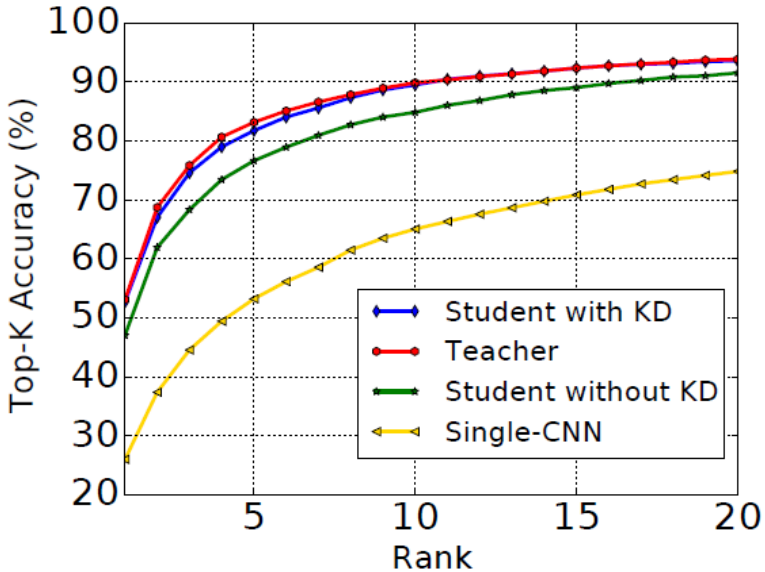
CNN architecture

- Use “data augmentation” to deal with issue of limited training data
- Measures similarity of query image with standard image of each pill-category in the CNN feature space. The similarity scores of each of the three CNNs are added together. $S_{final} = S_{color} + S_{gray} + S_{gradient}$
- Each CNN is based on AlexNet. They are similar except that for color CNN, which has 3 input channels (RGB). Other two CNNs have 1 channel.

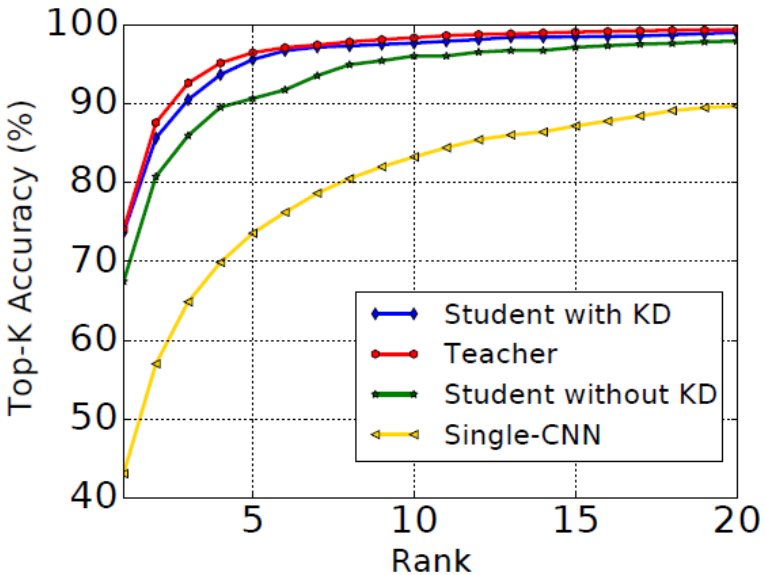
Model-compression using knowledge distillation

- Challenge: their system has 40M parameters and 3.2B FLOP computation
- Use “knowledge distillation” strategy to train a smaller student network
- They also apply other optimizations, such as using 1x1 filter for channel reduction.

Results



One side pill image recognition



Two side pill image recognition

Runtime in millisecond

	i7-5930k CPU	GTX 1080 GPU	TX1 GPU	Galaxy S7 CPU
Teacher	163	4.2	40	2407
Student	114	3.8	19	1335

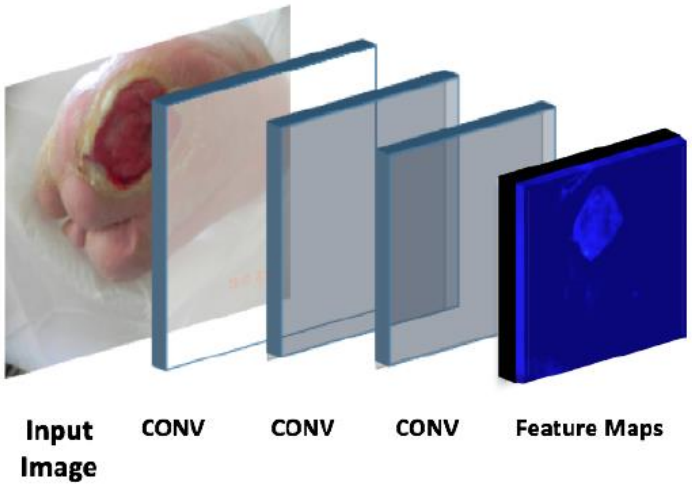
Detecting and localizing “diabetic foot ulcers” (DFU)

Goyal et al., “Robust Methods for Real-time Diabetic Foot Ulcer Detection and Localization on Mobile Devices, IEEE Journal of Biomedical and Health Informatics, 2018

Classification Localization Segmentation

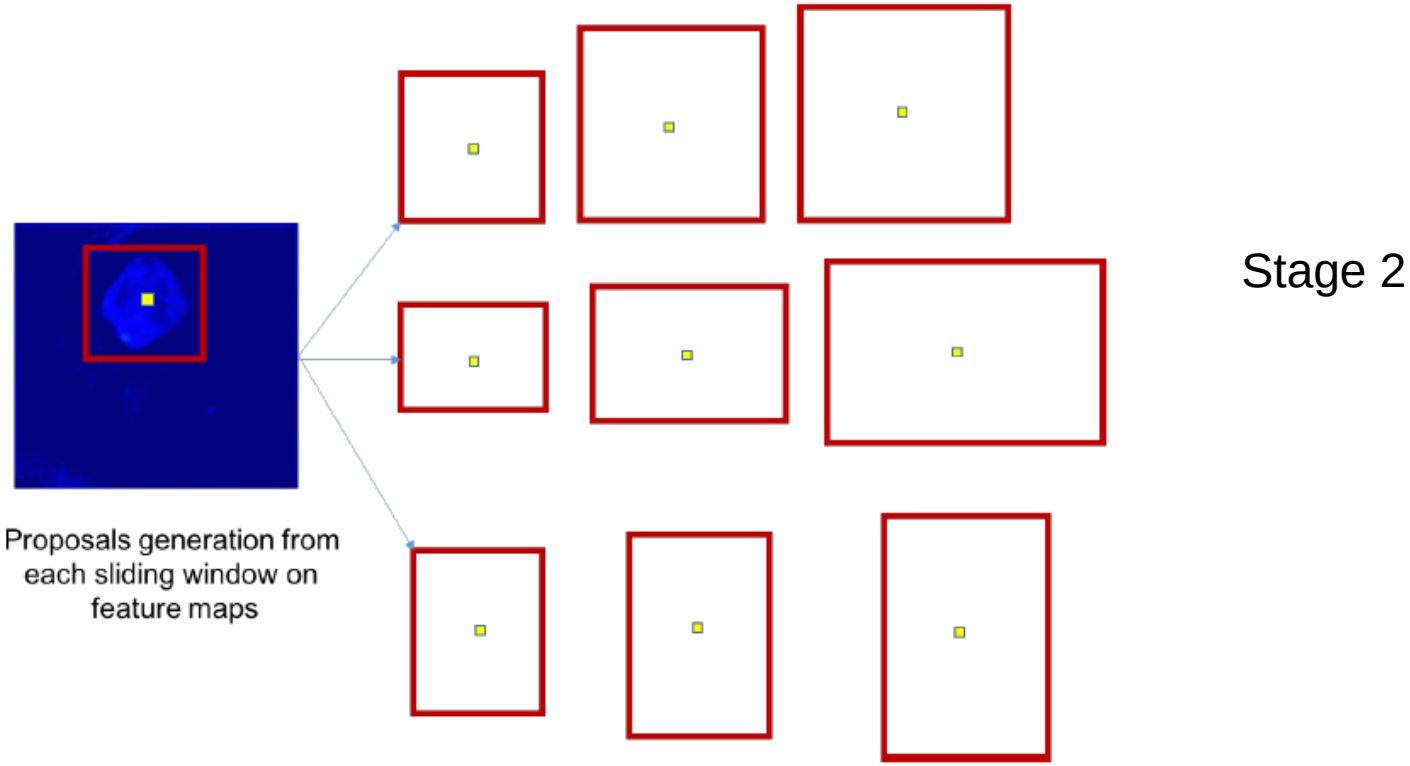
- They create a dataset. Annotate ground truth with help of experts in form of bounding box of DFU

1. Use a CNN for extracting features from input images.
2. From these, regions containing DFU are identified.
3. Finally, these regions of interest are classified (if more than one class) and their locations are further refined.



Stage 1

The feature map extracted by CNN that acts as backbone for object localization network



Detected proposal boxes with translate/scale operation to fit the object. There can be several proposals on a single object.

- For each of these steps, they evaluate various well-known CNN architectures.
- SSD-MobileNet uses MobileNet for classification and SSD for localization.

Performance of object localization models on MS-COCO dataset

Model Name	Latency *(ms)*	*Size of Model (MB)*	*COCO mAP*
SSD-MobileNet	30	29.2	21
SSD-InceptionV2	42	102.2	24
Faster R-CNN with InceptionV2	57.2	58	28
R-FCN with ResNet101	92	218.3	30
Faster R-CNN with ResNet101	106	196.9	32
Faster R-CNN with Inception-ResnetV2	620	247.5	37

Optimizations

- Medical dataset is limited
- => use both partial and full transfer learning approach to transfer the features from a model trained on other datasets such as ImageNet to their domain

Results on their DFU dataset

Results using TITAN X 12GB GPU

Model Name	Latency *(ms)*	*Size of Model (MB)*	*Ulcer mAP*	*Overlap Percentage (%)*
SSD-MobileNet	28	22.6	84.9	89.4
SSD-InceptionV2	37	53.5	87.2	92.6
Faster R-CNN with InceptionV2	48	52.2	91.8	95.8
R-FCN with Resnet 101	90	199.1	90.6	96.1

- On TX2: SSD-MobileNet takes 70ms

References

- S. Mittal, “A Survey on Optimized Implementation of Deep Learning Models on the NVIDIA Jetson Platform”, 2019 (link)